

Matin Ebrahimkhani

Curriculum Vitae

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RESEARCH INTERESTS

- Natural Language Processing:**
Large language models, language understanding and generation, text correction, sequence labeling, multilingual NLP
- Speech and Multimodal Language Systems:**
Automatic speech recognition, text-to-speech systems, neural voice generation, language-speech integration, and multimodal learning
- Efficient and Trustworthy AI:**
Model adaptation and fine-tuning, efficient training and inference, evaluation of AI systems, and robust machine learning
- Applied Machine Learning Systems:**
Scalable AI pipelines, real-world deployment of machine learning models, and reproducible AI research

EDUCATION

- 2020–2024 **M.Sc. in Computer Engineering – Artificial Intelligence and Robotics**
Department of Computer Science and Information Technology
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran
- GPA: 3.11/4 (15.54/20)**
 - Thesis: Correction of Space and ZWNJ in Persian Text, Using Sequence Labeling with Pre-trained Large Language Models**
 - Mentor: Dr. Ebrahim Ansari
 - Main Courses: DL, ANN, ML, Data Mining, Statistical Pattern Recognition
- 2016–2020 **B.Sc. in Information Technology Engineering**
Department of Computer Science and Information Technology
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran
- GPA: 3.25/4 (16.44/20)**
 - Thesis: Semi-Automated Forex Assistant**
 - Mentor: Dr. Ebrahim Ansari
 - Main Courses: ML, NLP, Statistical Models, Intelligent Systems Design

PUBLICATIONS AND CONFERENCES

PerSpaCor: Correcting Space and ZWNJ Errors in Persian Text with Transformer Models

Matin Ebrahimkhani, Ebrahim Ansari

Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing (RANLP 2025)

- Novel Research:** Engineered and evaluated four original transformer-based classifier architectures—ReLU-Norm, DualStep DropNet, and SimplexNet—achieving a **State-of-the-Art macro F1-score of 97.26%** (98.38% with interactive correction).
- Dataset Leadership:** Independently curated, preprocessed, and **manually annotated** a massive dataset of over **12.7 million words**, creating a new high-quality benchmark for Persian NLP.
- Open Source Impact:** Released all state of the art models, code, and the annotated corpus to the research community and deployed a public web-based correction system to make the research accessible to end-users.
- Paper:** ACL Anthology – 2025.ranlp-1.40
- Artifacts:** Hugging Face/PerSpaCor | GitHub/PerSpaCor_components

HONORS AND ACHIEVEMENTS

- 2024 **Ranked 58th Nationwide**
National Ph.D. Entrance Examination (Concours)
- 2022 **LLM Research Grant Recipient**
Awarded funding to support Large Language Model research
- 2021 **Founder of IASBS Music Society**
Established the student music community at the IASBS
- 2020 **Ranked 21st Nationwide**
National Graduate Entrance Examination (Concours)
M.Sc. admission in Computer Science
- 2019 **1st Place — IASBS ACM Programming Competition**
Member of the first competitive programming team at IASBS

SKILLS

Programming Languages

PYTHON	Expert — ML, NLP, and automation
C/C++	Advanced — systems and performance
BASH	Proficient — scripting and automation
JAVASCRIPT	Familiar — web development
HTML/CSS	Advanced — web design and development
SQL	Familiar — database operations
C#	Familiar — application development

Machine Learning and Deep Learning

PYTORCH	Advanced — research and production
TENSORFLOW	Intermediate — prototyping
HUGGING FACE	Advanced — transformers, fine-tuning Datasets and NLP pipelines
CUDA	Intermediate — GPU computing
TENSORBOARD	Intermediate — monitoring and analysis

Software Engineering and Tools

LINUX	Proficient — daily OS and sysadmin
GIT	Proficient — VC and collaboration
DOCKER	Proficient — containerization
KUBERNETES	Intermediate — service management
LaTeX	Expert — academic writing

Mathematical Foundations

MACHINE LEARNING	Modeling and evaluation
DEEP LEARNING	Neural architectures and optimization
ALGORITHMS	Design and analysis
OPTIMIZATION	Applied mathematical methods
PROBABILITY THEORY	Statistical modeling
LINEAR ALGEBRA	Mathematical foundations

Languages

ENGLISH	Near-native
PERSIAN	Native
AZERBAIJANI	Native
TURKISH	Elementary

Efficient Adaptation of Foundation Models: A Comparative Study of Parameter-Efficient Fine-Tuning Methods for Domain-Specific Tasks

Matin Ebrahimkhani, Mina Sirinchi, Mo Nasher, Mobin Ebrahimkhani
(In Progress)

Investigating parameter-efficient fine-tuning techniques (LoRA, QLoRA, Adapters, Prefix Tuning, and full fine-tuning) across multiple NLP tasks (text classification, question answering, sequence labeling, text generation). The study will evaluate accuracy/F1 (for classification/QA/labeling), BLEU/ROUGE (for generation), and resource metrics (training time/GPU memory, inference latency, and number of trained parameters) to compare performance–efficiency trade-offs. Planned code and scripts will be released for reproducibility.

RESEARCH AND WORK EXPERIENCE

REEICO
AI Researcher and Developer

Designed and developed an end-to-end **AI-powered dubbing system** for translating and localizing English video content into Persian. The pipeline integrated automatic speech recognition (ASR), **neural machine translation** (NMT), **text-to-speech synthesis** (TTS), and speaker-aware voice cloning. Implemented speaker diarization and alignment techniques. Independently handled the **MLOps** infrastructure, including model training pipelines, version control, containerization, and scalable deployment for reproducible and maintainable research.

Freelance
Freelance Developer and Researcher

Designed and implemented web-based services and AI-driven applications across various domains. Conducted independent research on reinforcement learning and algorithmic trading, developing custom trading algorithms and indicators for the Forex market. Emphasized real-time data processing, predictive modeling, and the integration of machine learning into financial decision-making systems.

SAFA Corp.
Software Engineer and Data Analyst

Developed semi-automated and fully-automated robots for the Forex market. Performed statistical analysis on financial markets. Designed and implemented machine learning algorithms on financial market data.

Naghil Noghouli
Programmer & Technical Lead

Led the technical development of Naghil Noghouli, an interactive storytelling game-app for children aged 3–12 built with C# and the Unity engine.

Chakavak Music Academy
Programmer

Implemented a learning management system (LMS) for Chakavak Music Academy, Zanjan, Iran.

OTHER INTERESTS

Free and Open-Source Software (FOSS), Computer Hardware and PC Building, DIY Home Projects, Canines, Percussion Instruments, Overlanding and Camping, Traveling, Cinema, Discovering Exotic Music, Poetry, and Fixing Things That Weren’t Technically Broken.

TEACHING ASSISTANT EXPERIENCE

Dept. of Computer Science and Information Technology, IASBS

WINTER 2022	Deep Learning Concept lectures, assignment design and grading
FALL 2021	Intro to Computer Science Problem-solving sessions and programming labs
SPRING 2020	Information Retrieval Written and programming assignment evaluation
WINTER 2020	Advanced Programming II Recitation sessions, lab design and grading
FALL 2019	Advanced Programming I Exercise sessions and introductory C++ labs
FALL 2018	Digital Logic Circuits Assignment evaluation and lab coursework grading

REFERENCES

Dr. Ebrahim Ansari
Assistant Professor, Dept. of Computer Science and Information Technology, IASBS, Zanjan, Iran
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Dr. Zahra Narimani
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Dr. Parvin Razzaghi
Assistant Professor, Dept. of Computer Science and Information Technology, IASBS, Zanjan, Iran
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